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Lesson Notes

MODULE 5 | LESSON 3

INTRODUCING RISK PARITY

Reading Time	15 minutes
Prior Knowledge	Traditional portfolio theory and utility theory.
Keywords	behavioral finance, bounded rationality, satisfice, prospect theory, cognitive biases, emotional biases.

In the first two lessons of this module, we studied the Kelly Criterion portfolio as a more aggressive (higher return and higher risk) approach to portfolio construction. In a sense, risk parity takes us in the other direction, sacrificing higher returns in favor of lower volatility. But the story does not end there: We also learn about an approach to enhance risk parity by modifying the constraints (relaxing one and adding another). This novel approach is especially satisfying because of the simplifying transformation it performs in order to solve its more complicated optimization problem.

As we review a growing number of portfolio construction approaches, it helps to compare and contrast the approaches, especially in the context of portfolio theory. The [first required reading](#) serves us well here: It reviews the development of portfolio optimization approaches, from mean-variance optimization (MVO) through the Kelly criterion portfolio (KCP) and up to risk parity (RP). The clear side-by-side analysis should be helpful. The circumstances under which KCP and RP collapse into MVO equivalents should also be insightful, so pay close attention to these derivations and the assumptions that make them possible.

The [second required reading](#) describes an interesting model that seeks to not only maintain the lower risk inherent to the RP portfolio but also improve the returns. This would seem to come at the cost of RP's stability: Traditional RP does not include an estimate of asset returns, and therefore, traditional RP is free of the sensitivity and estimation error associated with these estimates. However, the authors Gambeta and Kwon indicate that their proposed approach is still resilient to these issues. What's more, this relaxed risk parity (RRP) portfolio is also supposed to be more diversified than the MVO and KCP portfolios because the objective still includes parity across the risk contributions; the objective is simply relaxed to some degree in favor of slightly higher returns. The degree of relaxation or deviation from the strict risk parity constraints is defined and controlled by the user through the new parameter ρ .

In addition to innovative enhancement of a traditional portfolio construction, this article demonstrates another practice that we as financial engineers should learn to emulate: They take a very difficult and complex problem and reframe it in a form that can be solved by some simple or well-established method. Specifically, they take a fourth-order polynomial objective function, which is neither linear nor convex, and reformulate it “by transformation of the constraints into cone constraints and a transformation of the objective into linear form” making it “more friendly and flexible to additional constraints to deal with the expected return information” (Gambeta and Kwon 3).

This master stroke by Gambeta and Kwon incorporates second-order conic programming, which you will learn much more about by watching the [required “reading.”](#) In general, conic programming involves a convex optimization problem that optimizes a linear function over the intersection of affine spaces and convex cones. If this sounds daunting, the visual explanation in the video will help you considerably. It is worthwhile to learn about this application of second-order conic programming (SOCP) for two reasons:

- 1) Incorporating SOCP in this innovation is just another example of the expanding frontiers of financial engineering techniques by exploring adjacent subfields (in this case, conic programming).
- 2) Conic programming, and SOCP in particular, are growing in popularity as crucial tools for solving difficult problems because they offer closed-form solutions and neat duality.

In this lesson, we discovered risk parity in comparison to Kelly Criterion and mean-variance optimization. We also worked through a proposed enhancement of the risk parity approach, which entailed a conic programming technique, known as second-order conic programming (SOCP). In the next lesson, we continue to focus on risk parity as the foundation for portfolio construction, but we will explore even more sophisticated modifications of this portfolio optimization approach: Specifically, we apply machine learning techniques in our search for an improvement on this modern standard of portfolio construction.

References

- Gambeta, Vaughn, and Roy Kwon. "Risk Return Trade-Off in Relaxed Risk Parity Portfolio Optimization." *Journal of Risk and Financial Management*, vol. 13, no. 10, 2020, p. 1–28, <https://www.mdpi.com/1911-8074/13/10/237>
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- Sato, Yoshiharu. "Model-Free Reinforcement Learning for Financial Portfolios: A Brief Survey." *arXiv.org*, 2019, p. 1–20, <https://arxiv.org/abs/1904.04973>